Tesla demonstrated a radio-controlled boat, proving that wireless control of machines was possible, a breakthrough that later inspired aerial applications.

By the early 20th century, as aviation rapidly advanced following the Wright brothers' first flight (1903), engineers began to explore whether aircraft could be flown without onboard pilots. These early experiments culminated during World War I, when several nations invested in the creation of pilotless aircraft for military purposes. The "Aerial Target" (1916) developed by A. M. Low in Britain and the Kettering Bug (1918) in the United States were landmark projects in this period. The Aerial Target was one of the first radio-controlled aircraft, while the Kettering Bug used a simple gyroscopic guidance system and mechanical timer to deliver explosives over short distances. Although their operational success was limited, these prototypes proved the feasibility of unmanned flight and set the stage for future UAV development.

These early 20th-century experiments reflected the intersection of mechanical innovation, emerging radio technology, and military necessity. While constrained by the lack of reliable power sources and communication systems, they introduced critical ideas such as remote piloting, automated control, and aerial targeting that became the foundation for modern drones.

2.3.2 The role of balloons and kites in aerial surveillance

Long before the advent of powered flight, balloons and kites played an instru-mental role in the early history of aerial observation and unmanned flight concepts. As early as the 18th and 19th centuries, military strategists recog-nized the value of elevated platforms for reconnaissance and communica-tion. These early aerial systems, though not truly "unmanned aircraft" in the modern sense, served as precursors to UAVs by